

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATIONS  
Higher 2

**MARK SCHEME**

CANDIDATE  
NAME

CLASS

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**BIOLOGY**

Paper 2 Core Paper

**9747/02**

**September 2009**

**2 hours**

Additional Materials: Answer Paper

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**READ THESE INSTRUCTIONS FIRST**

Write your name and CT on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**

Answer **all** questions.

**Section B**

Answer any **one** question.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| <b>Section A</b>   |  |
| <b>1</b>           |  |
| <b>2</b>           |  |
| <b>3</b>           |  |
| <b>4</b>           |  |
| <b>5</b>           |  |
| <b>6</b>           |  |
| <b>7</b>           |  |
| <b>Section B</b>   |  |
| <b>Total</b>       |  |

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This document consists of **18** printed pages and **no** blank page.

**[Turn over**

## Section A

Answer **all** the questions in this section.

- 1 When present in water, carbon dioxide dissolves, forming carbonic acid, which then dissociates immediately to form bicarbonate ions and protons:

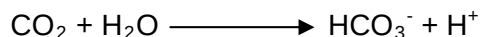


Fig. 1.1 shows the energy changes which take place during this reaction.

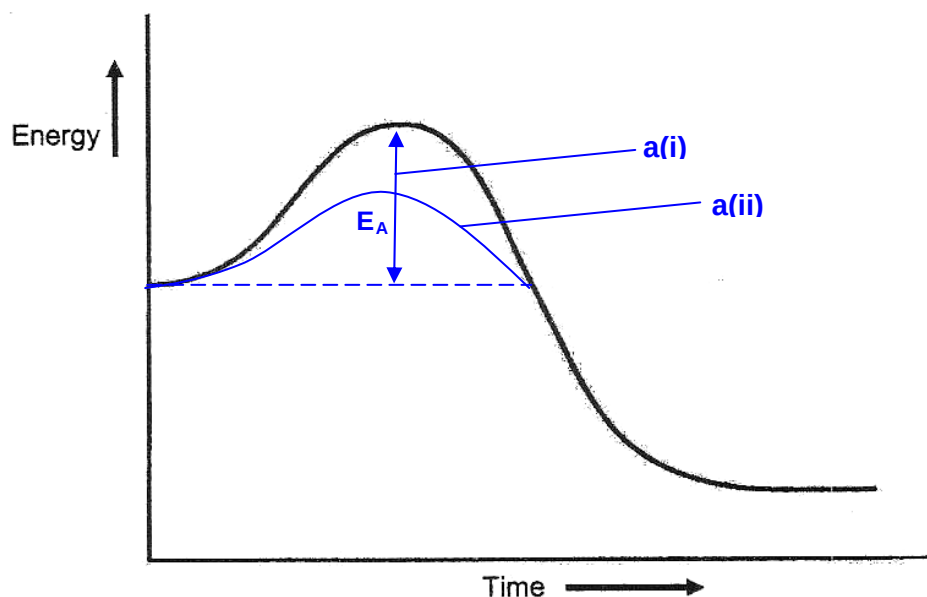


Fig. 1.1

The reaction also occurs in the human blood, where it is catalyzed by the enzyme carbonic anhydrase. The pH of blood is 7.4.

(a) On Fig. 1.1,

- (i) indicate the activation energy ( $E_a$ ) required for the uncatalyzed reaction; [1]
- (ii) sketch and label the energy changes which take place when carbonic anhydrase is present. [1]

(b) Explain how carbonic anhydrase brings about the energy changes in the enzyme-catalyzed reaction.

Allow close proximity of reactants due to temporary binding of substrates ;

Ensures correct orientation of reactants;

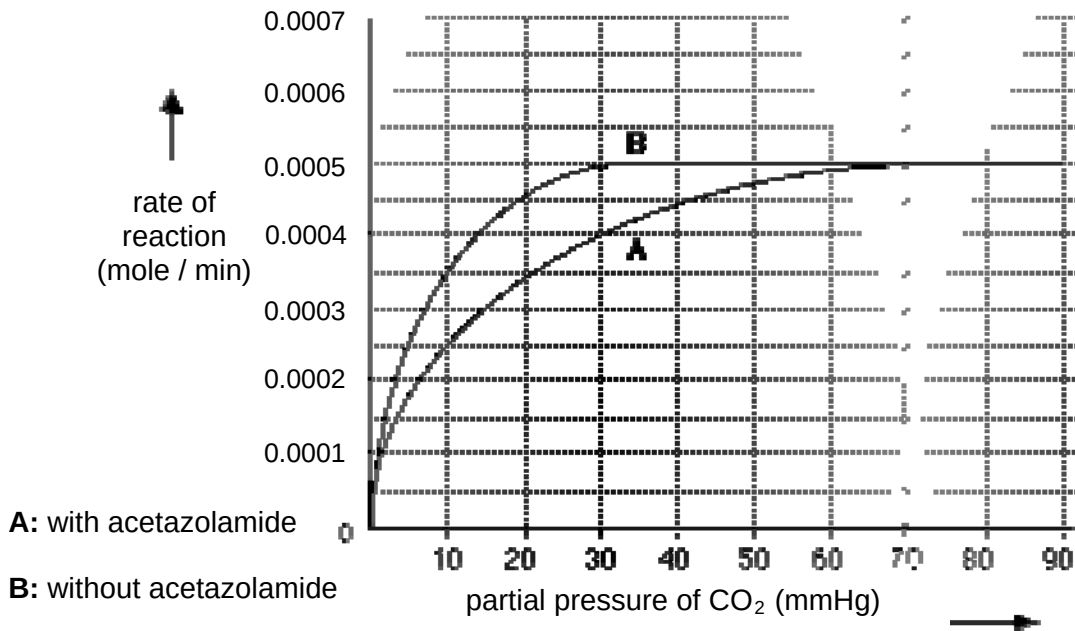
Destabilizing the bonds of reactants as enzymes distort reactants' bonds;

Active site creates a water free zone for non-polar reactants may react more easily;

R-groups of enzymes can change the charge / alter the distribution of electrons on the substrate;

[3]

- (c) **Fig. 1.2** shows the rate of reaction-partial pressure of  $\text{CO}_2$  curves for carbonic anhydrase. Graph **A** shows the curve in the presence of acetazolamide, which is an inhibitor to carbonic anhydrase.



**Fig. 1.2**

- (i) With reference to the curves in **Fig.1.2**, explain the mechanism by which acetazolamide inhibits carbonic anhydrase.

Competitive Inhibition as  $V_{\text{max}}$  can be reached at high  $\text{ppCO}_2$ ;

Competes with substrate for active site at low  $\text{ppCO}_2$ , thus rate of reaction is lower;

Inhibitor is structurally similar to that of substrate molecule / complementary to active site of enzyme;

[3]

- (ii) Suggest why acetazolamide is not broken down by carbonic anhydrase.

Catalytic residues / domains of enzyme unable to recognize / act on bonds of inhibitor;

[1]

- (d) It was found that increasing the concentration of carbonic anhydrase increased the rate of uninhibited reaction slightly but the rate returned to normal after a while. With reference to the given equation, suggest why this is so.

Increase in [enzyme] increases rate of E-S complex formation;

More  $\text{H}^+$  produced, causing change / lowering of pH;

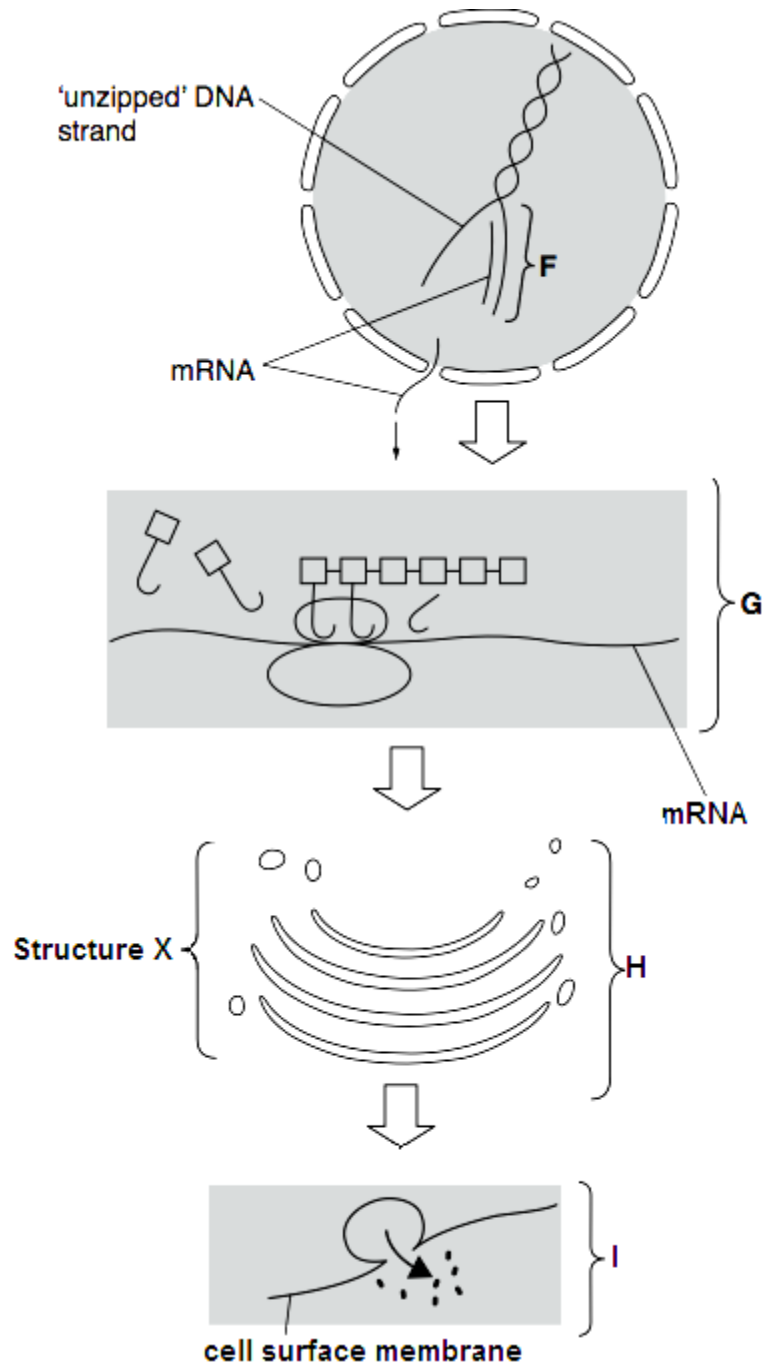
Change in charge of R-groups in active site decreases ability of substrate to bind;

® denaturation

[3]

**[Total: 12]**

- 2 **Fig. 2.1** is a diagram showing the sequence of events in the production and secretion of a hormone by a cell.



**Fig. 2.1**

- (a) Name the processes occurring at F, G, H and I.

*Any 2 points for 1 mark*

F Transcription

G Translation

H Post-translational modification @ any function of GA

I Exocytosis / secretion

[2]

- (b) Outline the events that occur during **F**, leading up to **G**.

During F, mRNA is synthesized by RNA polymerase using DNA as the template;

mRNA undergoes post-transcriptional modification / RNA processing and enter cytoplasm via nuclear pore;

Ribosomal subunits, aminoacyl-tRNA and mRNA associate to carry out translation at G;

[3]

- (c) Describe the function of structure **X**.

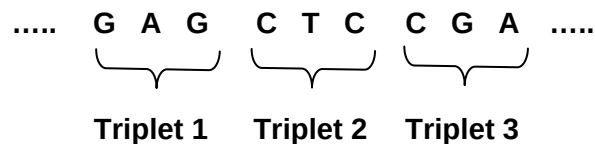
Chemical modification of hormone;

Temporary storage / Packaging hormone into vesicle;

Concentrates hormone molecules into vesicles;

[1]

- (d) Part of the DNA base sequence for this hormone is shown **Fig. 2.2**.



**Fig. 2.2**

- (i) In **Table 2.1**, state the messenger RNA (mRNA) codon and the transfer RNA (tRNA) anticodon that correspond to each of these DNA triplets. [1]

|                       | Triplet 1 | Triplet 2 | Triplet 3 |
|-----------------------|-----------|-----------|-----------|
| <b>mRNA codon</b>     | CUC       | GAG       | GCU       |
| <b>tRNA anticodon</b> | GAG       | CUC       | CGA       |

**Table 2.1**

- (ii) Explain the relationship between the sequence of the DNA triplets and the structure of the hormone.

Sequence of nucleotide on DNA determines sequence of amino acids in polypeptide;

Different R-group interactions result in unique three-dimensional structure of hormone;

[2]

**[Total: 9]**

- 3 (a) A virus is a microscopic infectious agent that can reproduce only inside a host cell.

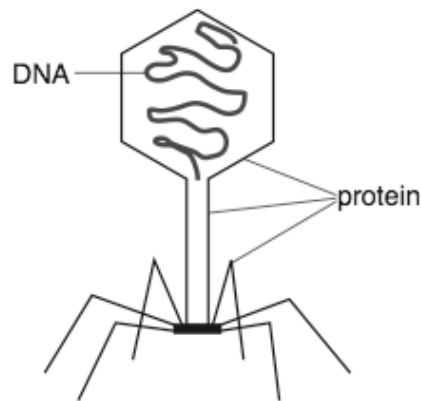


Fig. 3.1

**Fig. 3.1** shows a type of virus called a T2 phage. It consists of a protein coat molecule enclosing a DNA molecule. T2 phage has a reproductive cycle similar to T4 bacteriophage.

- (i) List **two** features of a bacterium that T2 phage does not share.

Cytoplasm / Cell membrane / Circular DNA / Bacterial cell wall / Ribosomes / Pili / Flagellum;;

[2]

- (ii) Discuss whether T2 phage is considered a living or non-living organism.

Considered to be living as it can reproduce to replicate more viruses / adapt to environment by exhibiting high mutation rates;

Considered to be non-living as it is not made up of cells / lacks organelles / enzymes needed for metabolism / does not carry out homeostasis / respond to stimuli;

Remains inconclusive as to whether it is living or non-living;

[3]

- (iii) Describe how the reproductive cycle of T2 phage differs from lambda phage.

| Feature                  | T2 phage                                                                     | Lambda phage                                                               |
|--------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Type of cycle            | Lytic                                                                        | Lytic and Lysogenic                                                        |
| Integration of phage DNA | Not integrated into bacterial chromosome                                     | Integrated into bacterial chromosome as prophage                           |
| Mode of propagation      | Phage DNA directs synthesis of new viral particles using host cell machinery | Phage DNA replicates each time host cell divides whilst in lysogenic cycle |
| Destruction of host cell | Lysis of host cell upon release of new viral particles                       | Remains dormant in host cell unless induced to transit into lytic cycle    |

[4]

- (b) Hepatitis is an inflammation of the liver which may be caused by viruses, alcohol and drugs. Hepatitis B and C viruses can cause chronic disease which results in death of liver cells. Liver cells infected by the hepatitis virus switch on a gene called *Fas*, which induces death of cells by self-destruction. Pioneering research in using gene therapy has produced a strikingly successful treatment for hepatitis in mice. This technique is known as RNA interference.

Double-stranded RNA molecules, 21 to 23 nucleotides long, were injected into mice with hepatitis. The sequence of this 'small interfering RNA' (siRNA) matched part of the *Fas* gene. Once in the liver cell, the two strands of the siRNA were separated so that one strand could bind to the *Fas* mRNA transcript. This affected the stability mRNA and considerably increased the survival of mice with hepatitis.

- (i) Describe **two** ways in which the structure of siRNA differs from mRNA.

1. siRNA is smaller than mRNA / siRNA have fewer nucleotides / siRNA complementary to part of the gene while mRNA complementary to entire gene;
2. siRNA is double stranded initially but separates while mRNA is single-stranded;

[2]

- (ii) Suggest how hepatitis virus causes disease in humans.

Hepatitis virus triggers synthesis of FAS protein which promotes / causes apoptosis and liver cells die;

Toxin build-up / toxicity due to decreased detoxification;

[2]

- (iii) Suggest how the siRNA therapy can increase survival of mice.

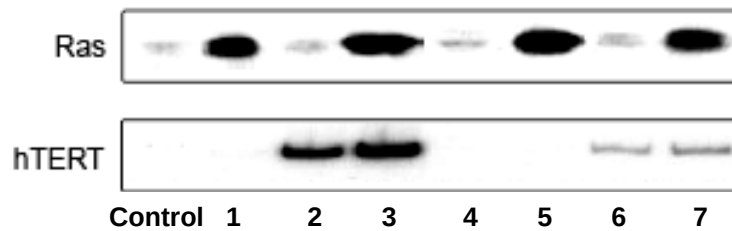
siRNA complementary to part of mRNA and binds, forming a RNA duplex / dsRNA, which is degraded;

FAS protein not translated / are absent, does not trigger apoptosis, so liver cells can carry out detoxification;

[2]

**[Total: 15]**

- 4 **Fig. 4.1** shows an analysis of Ras protein (signal transducer oncogene) and telomerase (hTERT) in seven cell types in a cancer patient. The control shows the amount of the two substances in a normal cell. A darkly stained band indicates that there is high amount of the respective substance in that particular cell type.



**Fig. 4.1**

(a) Using **Fig. 4.1**,

(i) state which cell type would have the highest chance of developing cancer;

3;

[1]

(ii) explain why the cell type in (i) could become cancerous.

Over-expression of Ras protein and active telomerase;

Due to gain-of-function mutation;

Cells divide too quickly and do not undergo apoptosis;

Resulting in a mass of cells known as tumour which causes cancer;

[4]



(b) Fig. 4.2 compares the effects of telomerase inhibitors in germ cells, stem cells and breast cancer cells.

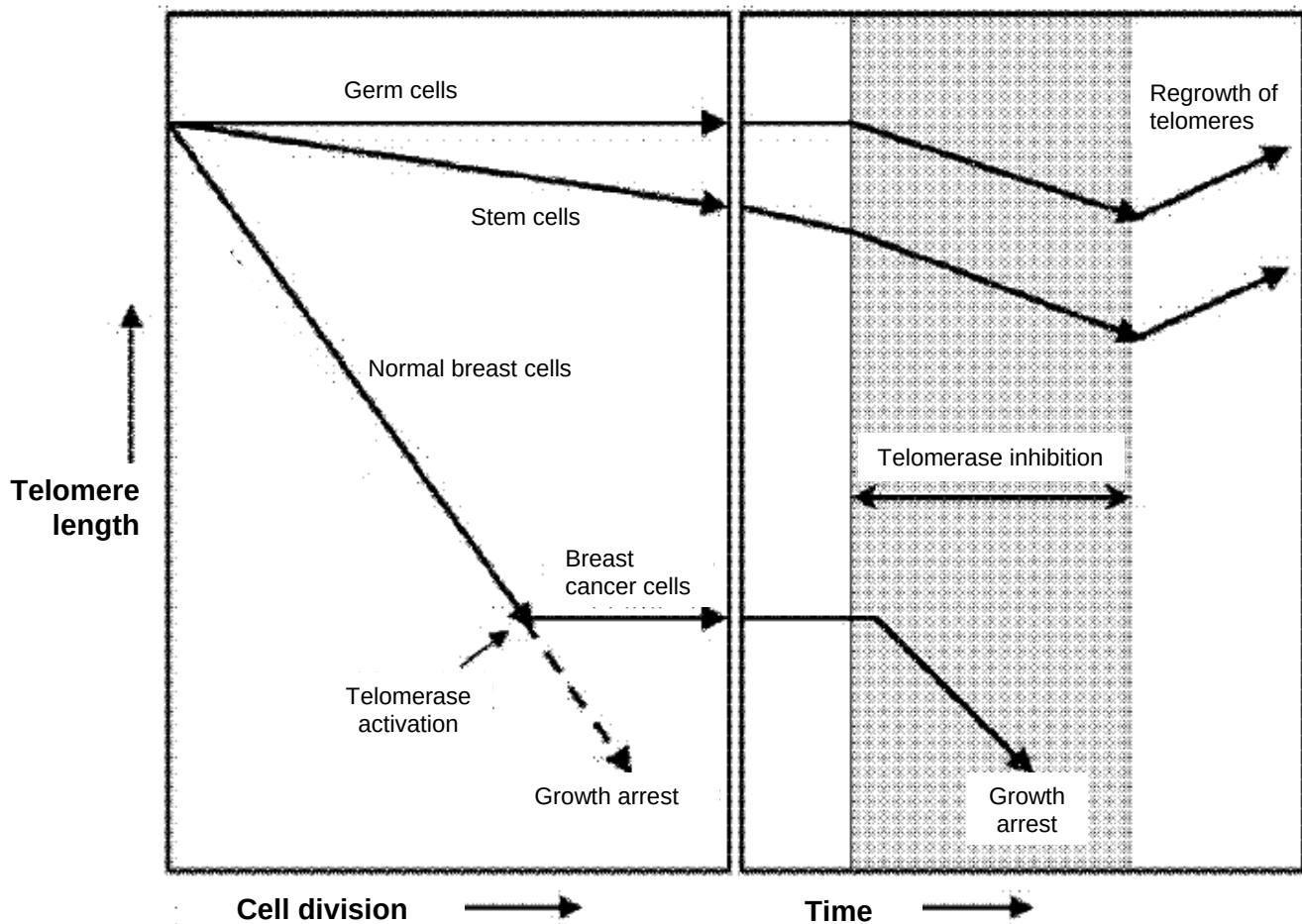


Fig. 4.2

With reference to Fig. 4.2,

(i) describe the effects of telomerase activation and telomerase inhibition on breast cells;

Telomerase activation causes length of telomeres to remain constant;

Leading to continuous division of breast cells;

Telomerase inhibition causes length of telomere to decrease;

Leading to growth arrest of breast cells;

[3]

(ii) suggest and explain if the use of telomerase inhibitors in breast cancer therapy is feasible.

Feasible because growth / division of breast cancer cells is arrested;

Growth of telomere length in germ and stem cells resumes once inhibition is removed;

[2]

[Total: 10]

- 5 Dopamine is a chemical naturally produced in the body. In the brain, dopamine functions as a neurotransmitter, activating dopamine receptors on the post-synaptic membrane. Dopamine concentration affects brain processes that control body movement, emotional response and the ability to experience pleasure and pain.

- (a) Describe the events that take place after the arrival of the pre-synaptic action potential that results in the activation of dopamine receptors on the post-synaptic membrane.

Depolarisation of pre-synaptic membrane causes voltage-gated calcium ion channels to open, resulting in  $\text{Ca}^{2+}$  influx;

Synaptic vesicles containing dopamine move along cytoskeleton towards pre-synaptic membrane to release dopamine via exocytosis;

Dopamine diffuses across synaptic cleft and binds to dopamine receptors on post-synaptic membrane;

[3]

- (b) State **two** reasons why nerve impulses can only travel in one direction across a synapse.

Dopamine is only found in the pre-synaptic neurone;

Dopamine receptors are found only on the post-synaptic membrane;

[2]

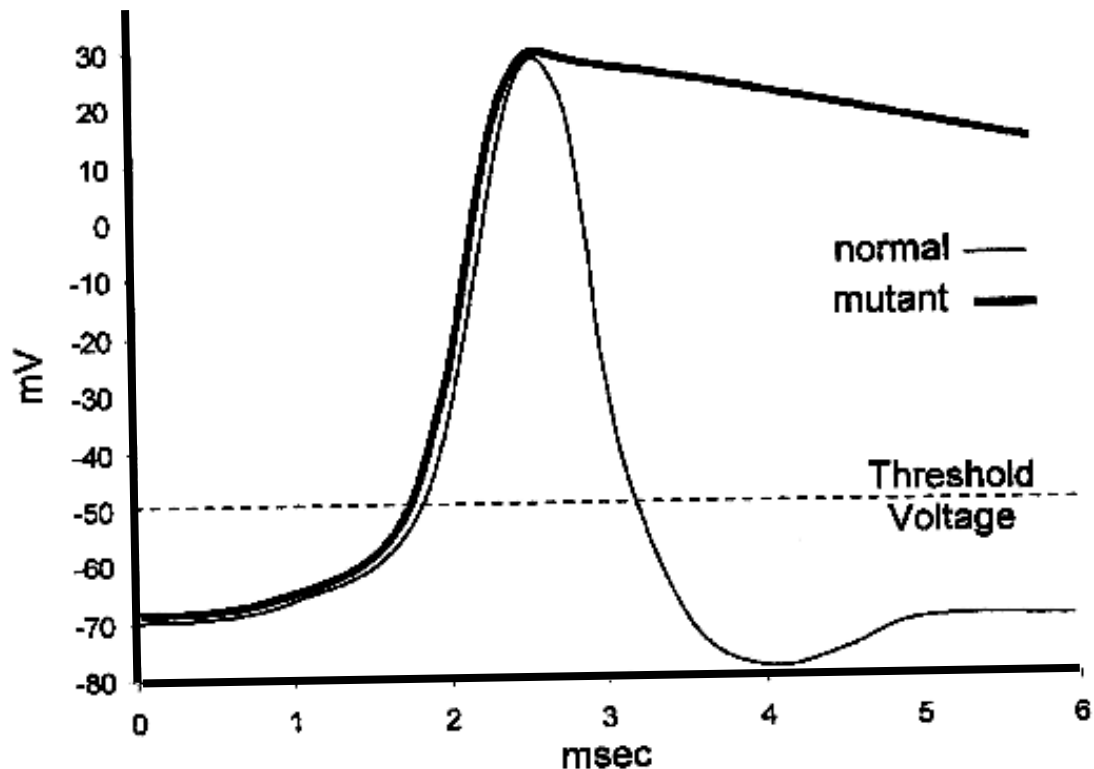
- (c) Suggest why the pre-synaptic neurone has an abundance of mitochondria in its synaptic knob.

Mitochondria synthesizes ATP during aerobic respiration;

ATP needed to actively pump  $\text{Ca}^{2+}$  out of pre-synaptic neurone / movement of vesicles towards cell surface membrane / synthesis of acetylcholine from acetate and choline;

[2]

**Fig. 5.2** shows the action potentials of a normal and mutant neurone. In the mutant neurone, a mutation in the gene coding for a channel protein results in a change to the action potential.



**Fig. 5.2**

**(d)** Describe how the resting potential of the normal neurone is maintained at -70mV.

$\text{Na}^+\text{-K}^+$  pump actively transports 3  $\text{Na}^+$  out and 2  $\text{K}^+$  in;

Facilitated diffusion by  $\text{Na}^+$  and  $\text{K}^+$  channels, which causes membrane to be more permeable to  $\text{K}^+$  / impermeable to  $\text{Na}^+$ ;

Impermeability of membrane to organic anions, which has a higher concentration inside neurone;

[2]

**(e)** With reference to **Fig. 5.2**, explain how the mutation results in a change to the action potential.

Repolarisation of membrane takes a longer time;

Mutation in the gene coding for  $\text{K}^+$  voltage-gated ion channels results in a non-functional channel;

Thus  $\text{K}^+$  cannot move rapidly out of the neurone;

[2]

**[Total: 11]**

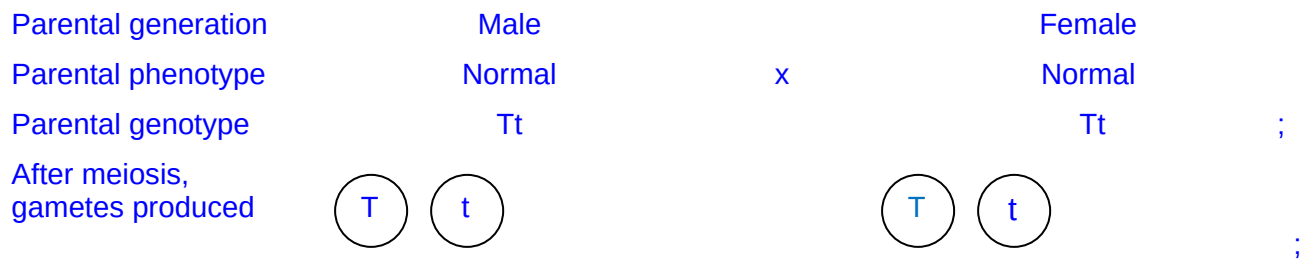
- 6 Tay-Sachs is an inherited condition caused by a recessive allele for the gene coding for an enzyme required in the regulation of the synthesis of glycolipids. Absence of this enzyme results in the accumulation of glycolipids in the central nervous system of affected individuals. This causes severe neurological symptoms within six months of birth and death of the individual between three and six years of age.

- (a) Suggest why individuals that are heterozygous for Tay-Sachs condition are apparently unaffected.

Possess one copy of the wild-type / dominant / normal allele which could still code for functional enzyme;

[1]

- (b) Draw a genetic diagram to show how two apparently normal parents may have an affected child.



By random fertilization of gametes,

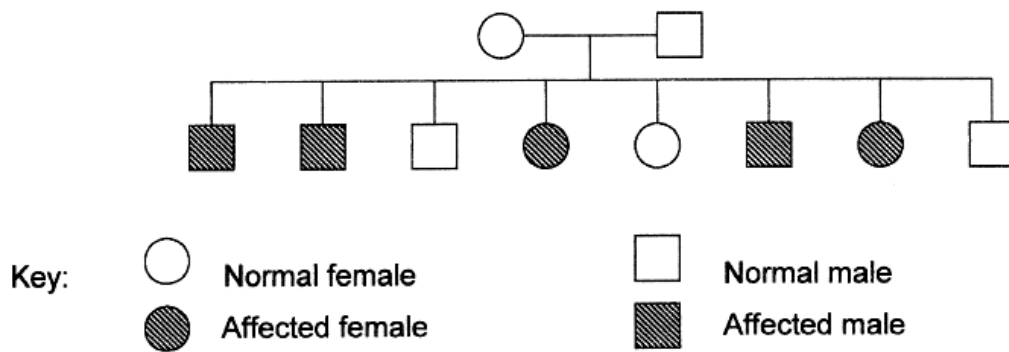
|                                                                                                        |                                                                                                        |                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--|
|                                                                                                        | <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block;">T</div> | <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block;">t</div> |  |
| <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block;">T</div> | TT                                                                                                     | Tt                                                                                                     |  |
| <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block;">t</div> | Tt                                                                                                     | tt                                                                                                     |  |

|                     |                                                               |    |    |            |   |
|---------------------|---------------------------------------------------------------|----|----|------------|---|
| F1 genotype         | TT                                                            | Tt | Tt | tt         |   |
| F1 phenotypic ratio | <div style="border-top: 1px solid black; width: 100%;"></div> |    |    | 1 Affected | ; |

3 Normal :

[4]

**Fig. 6.1** shows two generations of a family in which the Tay-Sachs condition is inherited.



**Fig. 6.1**

- (c) Determine, using an appropriate statistical test, if the observed results are expected. The critical values for the chi-squared test are provided in **Table 6.1**.

| Degrees of freedom | Probability |      |      |      |      |       |       |
|--------------------|-------------|------|------|------|------|-------|-------|
|                    | 0.50        | 0.20 | 0.10 | 0.05 | 0.02 | 0.01  | 0.001 |
| 1                  | 0.46        | 1.64 | 2.71 | 3.84 | 5.41 | 6.64  | 10.83 |
| 2                  | 1.39        | 3.22 | 4.61 | 5.99 | 7.82 | 9.21  | 13.82 |
| 3                  | 2.37        | 4.64 | 6.25 | 7.82 | 9.84 | 11.35 | 16.27 |

**Table 6.1**

| Phenotype | O | E | O-E | (O-E) <sup>2</sup> | (O-E) <sup>2</sup> /E |
|-----------|---|---|-----|--------------------|-----------------------|
| Normal    | 3 | 6 | -3  | 9                  | 1.5                   |
| Affected  | 5 | 2 | 3   | 9                  | 4.5                   |
|           |   |   |     | $\chi^2 = 6.0$ ;   |                       |

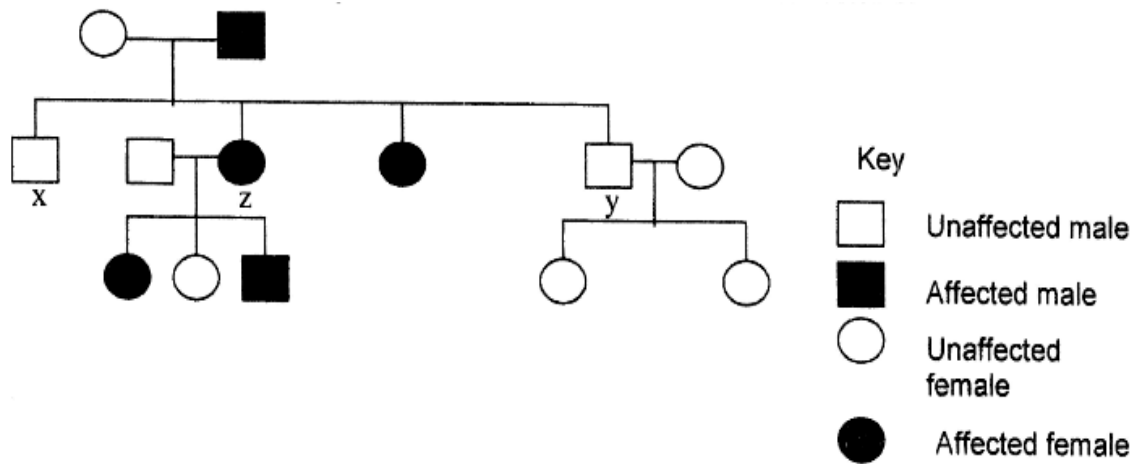
Degree of freedom = 2 - 1 = 1

Probability set at 0.05

Critical value = 3.84;

Since calculated  $\chi^2$  value is larger than critical value, there is significant difference between the observed and expected results and the difference is not due to chance but other factors;

Mental retardation is believed to be a sex-linked disease that is caused by the lack of production of an enzyme that participates in purine metabolism. **Fig. 6.2** shows a pedigree studying the inheritance of the disease.



**Fig. 6.2**

- (d) Explain using evidence from the pedigree shown in **Fig. 6.2**, whether the allele for the disease is recessive or dominant.

Dominant;

One of the daughters of female Z suffers from the disease;

Indicating that she must have inherited the allele from her mother since her father is normal;

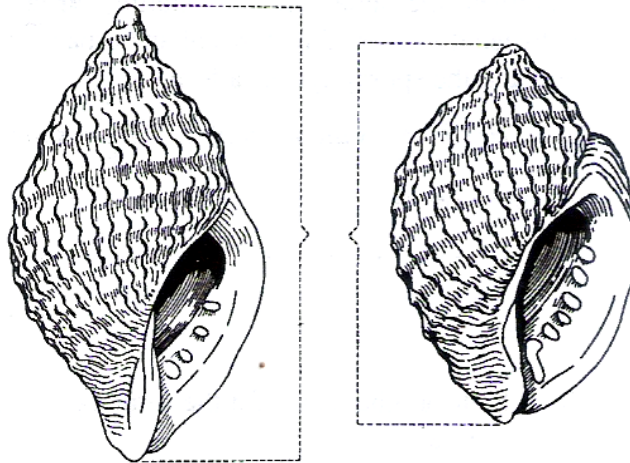
[3]

[Total: 11]

- 7 (a) The dog whelk (*Nucella lapillus*) is commonly found on rocky sea shores around the coasts of the British Isles. Considerable variability of its shell thickness and shape occurs on shores exposed to different strengths of wave action, as shown in **Fig. 7.1**.

On sheltered shores they have a high, thick shell and a relatively small aperture for attachment to rocks; on exposed shores the shell is thinner and stubbier, so that they are better able to hold on to the rocks they inhabit when subjected to the pounding of waves. The main predators of dog whelks are crabs, which will be active almost entirely on sheltered shores. Shell thickness is clearly protective here.

DOG WHELK



from a sheltered shore

from an exposed shore

Fig 7.1

Describe the role of natural selection, in maintaining the shell shape and thickness of populations of dog whelks on

- (i) sheltered rocky shores;

Variation in thickness of shells of dog whelks,;

Predation from crab exerts selection pressure on sheltered shores;

Whelks with thick shells are selected for (while thin shells are selected against);

Likes produce like, thus population of thick shelled whelks pass their desirable traits to offsprings (award only once in (i) or (ii));

[3]

- (ii) exposed rocky shores.

Variation in shell shape / aperture size of dog whelks,

Pounding waves act as selection pressure on exposed rocky shores;

Thin and stubby-shelled whelks / whelks with smaller aperture will be selected for as they are not washed away (while thick-shelled whelks / whelks with small aperture for attachment to rocks are selected against);

Likes produce like, thus population of thin shelled whelks pass their desirable traits to offsprings (award only once in (i) or (ii));

[3]

- (b) According to the Red Queen hypothesis, sexual reproduction persists because it enables many species to rapidly evolve new genetic defenses against parasites that attempt to live off them.

Scientists have tested this idea by observing different groups of small fish *Poeciliopsis* species (Gila topminnow) in Mexico. Some populations of the topminnow reproduce sexually, while others practice parthenogenesis. Parthenogenesis occurs when females produce offspring without any male contribution and the female's gametes develop directly into female offspring.

Topminnows are constantly parasitized by 'black spot disease' caused by black spot worms that encyst in the skin. Parasitized topminnows rarely survive. The researchers found that identical populations of the asexually reproducing topminnows harbored many more black-spot worms than did those producing sexually, a finding that fit the Red Queen hypothesis: the sexual topminnows could devise new defenses faster by recombination than the asexually producing clones.

However, it was observed that after a drought, the sexually reproducing topminnows were more heavily parasitized than the cloned topminnows. This process of whereby chance events cause the allele frequency to change unpredictably is known as genetic drift.

- (i) Name the evolutionary event that resulted in this genetic drift.

Bottleneck effect;

[1]

- (ii) Explain why there was an increase in the number of parasitized sexually reproducing topminnows after the drought.

Small population of survivors so lack genetic variation / resistant genes may be lost;

Reduced gene pool decreases chances of inheriting resistant genes to future generations;

[2]

The Neutral Theory of Molecular Evolution proposed that frequencies of alleles are not affected by natural selection but may increase or decrease as a result of genetic drift.

- (iii) Briefly describe the Neutral Theory of Molecular Evolution.

Genetic variation is due to accumulation of neutral mutations;

Neutral mutations do not affect phenotype of organism / do not confer any selective advantage or disadvantage to organism;

Arise due to the degeneracy / redundancy of genetic code / Wobble (base) effect;

'Survival of the Luckiest' in contrast with Darwin's 'Survival of the Fittest';

[3]

[Total: 12]



## Section B

Answer **one** question.

- 8 (a) Explain the significance of interphase in the cell cycle. [5]

1. Interphase occurs before M phase;
2. In G<sub>1</sub> phase, synthesis of new organelles occurs;
3. allowing the amount of organelles in daughter cells to be similar as parent cell;
4. Nucleolus actively synthesizes ribosomal RNA;
5. For synthesis of ribosomal subunits for subsequent translation of proteins;
6. In synthesis phase, DNA semi-conservative replication occurs;
7. DNA content doubles so diploid state is maintained;
8. Histone proteins are synthesized;
9. For condensation of chromatin into discrete units, chromosomes;
10. In G<sub>2</sub> phase centrosomes / centrioles replicates to form two centrosomes / centrioles;
11. For organization of spindle fibres in subsequent separation of chromosomes;

- (b) Outline the behaviour of chromosomes during meiosis. [9]

Prophase I

1. Chromatin fibres become more tightly coiled / condensed into chromosomes / nucleolus disperses / seems to disappear;
2. Homologous chromosomes pair to form bivalents / chiasmata formation which could result in crossing over;

Metaphase I

3. Each homologue is attached to kinetochore microtubules from centrioles at each pole;
4. Bivalents arrange themselves at the metaphase plate / arrangement of bivalent is independent of other bivalents / independent assortment of bivalent occurs;

Anaphase I

5. Each homologue of a bivalent separates towards opposite poles with centromere leading the way;

Telophase I

6. Chromosomes become less condensed with reappearance of nucleolus;

Prophase II

7. Chromatin fibres become more tightly coiled / condensed into chromosomes / nucleolus disperses / seems to disappear; (*award only once in either Prophase I or Prophase II*)

Metaphase II

8. Each chromosome is attached to kinetochore microtubules from centrioles at each pole;
9. Centromeres of chromosomes are positioned along metaphase plate / perpendicular to metaphase plate in metaphase I;

Anaphase II

10. Sister chromatids / Centromeres of each chromosome separates towards opposite poles with centromere leading the way;

Telophase II

11. Daughter chromosomes become less condensed with reappearance of nucleolus; (*award only once in either Telophase I or Telophase II*)

**(c)** Describe the development of cancer as a multi-step process. [6]

1. Accumulation of germ-line mutations / genetic changes inherited from parents / sporadic / somatic mutations which are acquired due to exposure to mutagens;
2. Loss-of-function mutation of both tumour suppressor alleles;
3. Gain-of-function mutation of at least one copy of proto-oncogene;
4. Activation of genes coding for telomerase in somatic cells;
5. Removes a natural limit on the number of cell divisions / results in uncontrolled proliferations of cells forming tumour;
6. Invasion of cancer cells into surrounding tissues;
7. Tumour angiogenesis allows cancer cells to enter the circulatory system;
8. Metastasis of cancer cells to distant sites resulting in formation of secondary tumours;

9 (b) Explain the eukaryotic processing of pre-mRNA. [6]

1. Capping changes 5' end of mRNA to a 3' end, by 5'-5' linkage / with the addition of 7-methylguanosine group;
2. Protects mRNA from degradation by 5' exonucleases / confers stability / assists in ribosomal binding;
3. Splicing removes introns and ligates exons to produce a mature mRNA;
4. Alternative splicing may occur, resulting in different combinations mature mRNA from a gene;
5. Addition of poly-A tail to 3' end of mRNA;
6. Allows slower degradation by 3' exonucleases / longer poly(A) tail implies longer half-life of mRNA / facilitates transport of mRNA from nucleus to cytoplasm;

(c) Compare and contrast between prokaryotic and eukaryotic control of gene expression at the translational and post-translation level. [9]

| Feature                  | Eukaryotic                                                                                                     | Prokaryotic                                                                             |
|--------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Initiation               | 1. Initiated by recognition of AUG sites by small ribosomal subunit;                                           |                                                                                         |
|                          | 2. First amino acid methionine                                                                                 | First amino acid N-formyl-methionine;                                                   |
| Translational repressors | 3. Translational repressors bind at 5' UTR to prevent translation initiation;                                  |                                                                                         |
| Stability of mRNA        | 4. Eukaryotic mRNA more stable / present in cytoplasm for longer period of time, due to 5' cap and poly A tail | Prokaryotic mRNA has no poly A tail or 5' cap, so less stable / have shorter half-life; |
| Antisense RNA            | 5. Antisense RNA produced prevent translation → duplex RNA which ribosome cannot bind / is degraded;           |                                                                                         |
| Biochemical Modification | 6. Addition of biochemical groups to proteins → activate protein;                                              |                                                                                         |
|                          |                                                                                                                | 7. Biochemical modification not as significant due to lack organelles such as ER & GA;  |
| Protein degradation      | 8. Unwanted / misfolded proteins can be degraded by proteasomes / lysosomal degradation                        | Lack lysosomes / proteasomes but possess other mechanism of degrading proteins;         |
| Feedback inhibition      | 9. Feedback mechanism present to control gene expression;                                                      |                                                                                         |

**(d)** Describe the significance of gene amplification. [5]

1. Definition – selective increase in number of copies of a particular gene without a proportional increase in other genes;
2. Meet the needs of cells resulting in higher level of mRNA and polypeptide synthesis at different development stages of cells;
3. Cause diseases like cancer due to over-expression of proteins leading to development of malignant tumours ;
4. Confer drug resistance, ref to example of methotrexate;
5. Confer selective advantage which allow organisms to survive in a particular environment;
6. Contribute to evolution of genome, ref to homologous genes;